

Condition-Based Maintenance for Grid Stations

Risk-Based Asset Strategy, Monitoring Data, Maintenance Triggers, and Practical Decision Rules

Prepared, reviewed, and issued by CESC

Website publication edition

Executive abstract

This paper explains how utilities and industrial owners can transition from calendar-based maintenance to risk-based condition maintenance across transformers, switchgear, protection systems, batteries, and auxiliary equipment.

The objective is not to add monitoring for its own sake, but to translate measured condition, duty history, and criticality into maintenance decisions that improve availability and reduce life-cycle cost.

Illustrative engineering examples are included to show how loading, gas generation trend, breaker duty, battery autonomy, and asset criticality can be converted into practical maintenance triggers.

Document basis

Document type	Technical white paper
Primary basis	IEC 61936-1, IEC 60599, IEC 62271 series, IEC 61850...
Intended audience	Utilities, EPC contractors, industrial asset owners, and engineering teams
Publication note	Prepared in formal issuer style to demonstrate CESC engineering capability in studies, protection, digital integration, retrofit, and asset management.

Applicable references

Reference	Application in this paper
IEC 61936-1	Power installation engineering and general station design context.
IEC 60599	Interpretation guidance for dissolved gases in transformer oil.
IEC 62271 series	Switchgear and breaker maintenance / condition context.
IEC 61850	Reference framework relevant to the engineering subject of this paper.
IEEE maintenance guidance where applicable	Reference framework relevant to the engineering subject of this paper.
IEC 61869-2	Current-transformer performance context relevant to monitoring accuracy and protection-supporting measurements.
IEEE C57.104	Supplementary transformer oil and dissolved-gas interpretation guidance where owner practice uses IEEE criteria.

Maintenance strategy shift

A fixed-interval maintenance model can either over-maintain healthy assets or miss fast-developing defects between inspection periods. Condition-based maintenance aims to focus effort where risk is rising, using measured condition, duty history, and consequence of failure.

For grid stations, the transition should be systematic rather than sensor-led. The question is not which devices can report data, but which asset states materially change maintenance decisions.

A sound engineering sequence is to define asset criticality, identify the dominant failure modes, assign measurable indicators, and then set trigger levels for inspection, maintenance deferral, repair, or replacement. This keeps the program aligned with real station risk rather than with generic maintenance checklists.

For example, a transformer at a primary bulk-supply station with no firm N-1 backup may justify a lower intervention threshold than the same transformer installed in a lightly loaded industrial station with standby capacity. The measured condition may be similar, but the allowable risk is different because the consequence of failure is materially different.

A simplified loading check can be used as an initial screen. If a 20 MVA, 66/11 kV transformer is operating at 16 MVA average load, the loading ratio is $16/20 = 0.80$ pu. If short-duration evening peaks reach 22 MVA, the peak loading ratio becomes $22/20 = 1.10$ pu. Repeated operation above rated loading does not automatically demand outage, but it should increase review of top-oil temperature, winding hot-spot margin, insulation ageing exposure, and cooling system performance.

For transformers, a practical engineering rule is to review both absolute values and rate of change. A hydrogen value of 120 ppm may be less concerning if stable for many months than a rise from 20 ppm to 120 ppm in a short period. Trending therefore matters as much as the latest reading.

Illustrative gas trend example: if dissolved hydrogen rises from 80 ppm to 140 ppm over 60 days, the average rise rate is $(140 - 80) / 60 = 1.0$ ppm/day. If the same asset historically increased by only 0.1 ppm/day, the condition review should move from routine watch to engineering investigation even before absolute alarm level is reached.

For circuit breakers, condition assessment should include fault interruption duty rather than operation count alone. A breaker that has made 2,000 normal operations and interrupted little fault current may be in better condition than a breaker with far fewer operations but several high-duty fault clearances. Duty-based maintenance therefore gives a more realistic view of wear.

A practical scoring method can assign likelihood on a scale of 1 to 5 and consequence on a scale of 1 to 5, giving a risk index from 1 to 25. This is sufficiently simple for field use while still allowing portfolio-level prioritization.

Illustrative example: a transformer showing accelerating gas trend, moisture increase, and repeated cooling alarms may be assigned likelihood = 4. If installed at a grid station where its failure would interrupt 18 MW of customer demand and no immediate standby unit exists, consequence may be assigned = 5. The resulting risk index is $4 \times 5 = 20$, which would normally justify planned outage and diagnostic action rather than continued routine observation.

By contrast, a feeder breaker with moderate mechanism timing drift but available redundancy may be assigned likelihood = 3 and consequence = 2, giving a risk index of 6. This may justify short-term monitoring and scheduled maintenance in the next window rather than urgent withdrawal from service.

Data governance is essential. Sensor drift, missing timestamps, and inconsistent naming can produce false alarms or hide emerging defects. A condition-based maintenance program should therefore define data validation rules, review intervals, escalation ownership, and decision records so that maintenance actions remain traceable.

A useful practice is to combine continuous indicators with periodic confirmatory tests. For instance, online temperature or dissolved gas trends may trigger targeted offline testing such as insulation resistance, tan-delta review, breaker timing measurement, or oil laboratory analysis. This layered approach avoids both overreaction to bad data and underreaction to genuine degradation.

In practical terms, this means CESC can convert broad maintenance objectives into engineering deliverables such as asset criticality registers, trigger matrices, monitoring architecture, alarm rationalization rules, diagnostic test plans, and brownfield implementation roadmaps for operating stations.

CESC can also support owner teams in linking maintenance data with protection, SCADA, and digital integration so that measured condition becomes decision-ready information rather than isolated data streams. This value addition is particularly important in retrofit projects where data quality, legacy equipment, and outage windows are constrained.

For condition-based maintenance, the engineering value lies in selecting the right indicators, checking them against equipment duty and failure consequence, and converting the result into action thresholds that are practical for station operation. This is where structured engineering adds measurable value beyond routine maintenance practice.

CESC's role is therefore not limited to preparing reports. It extends to defining condition logic, asset prioritization, intervention rules, retrofit pathways, and implementation methods that utilities and industrial owners can apply in real operating environments with budget, outage, and reliability constraints.

Condition-based maintenance should also feed the capital planning process. Assets that repeatedly return to intervention-planning state may no longer be good candidates for continued repair. The maintenance system should therefore support a repair-versus-replacement decision, especially where spare availability, outage consequence, or obsolescence risk is worsening.

The value of this framework is that it converts scattered maintenance observations into repeatable engineering decisions. It also creates a defensible basis for outage requests, spare-part planning, and management reporting.

Illustrative breaker duty example: if a 33 kV breaker is rated 25 kA short-circuit interruption and records three fault interruptions of 18 kA, 20 kA, and 22 kA, the cumulative normalized duty index can be screened as $(18/25)^2 + (20/25)^2 + (22/25)^2 = 0.52 + 0.64 + 0.77 = 1.93$. Even though this is not a direct replacement for manufacturer wear curves, it gives the maintenance engineer a practical indicator that the breaker has seen meaningful interruption stress and warrants timing, contact, and mechanism review.

For station batteries, autonomy should be checked against essential DC load rather than against battery age alone. If the emergency DC load is 28 A and the minimum required autonomy is 8 hours, the basic demand is $28 \times 8 = 224$ Ah. With ageing factor and design margin, the installed bank may need to remain materially above this effective requirement to avoid a hidden loss of station reliability.

Asset groups and indicators

Typical station asset groups include power transformers, switchgear and breakers, instrument transformers, protection and control panels, batteries and chargers, AC/DC auxiliaries, and communication infrastructure.

Each group needs a tailored indicator set. For example, transformer maintenance may rely on gases, moisture, and temperatures, while breaker maintenance may rely more on operation count, mechanism timing, contact wear, and gas or vacuum condition indicators.

Decision framework

A practical condition-based program should define normal state, watch state, intervention planning state, and urgent action state. Trigger thresholds should be linked to maintenance actions so that the system produces decisions rather than noise.

A simple risk expression can be framed as $\text{risk} = \text{likelihood} \times \text{consequence}$. This helps prioritize intervention where defect probability and outage impact are both significant.

Execution and governance

Condition-based maintenance requires data quality rules, ownership of review, alarm rationalization, and a closed loop between operations, maintenance, and asset management. Without governance, the maintenance team receives many signals but little usable guidance.

It is also important to preserve engineering judgment. Trending and analytics should support experienced review, not replace it.

CESC engineering contribution

CESC can define the station maintenance framework, select practical indicators, structure trigger matrices, integrate monitoring outputs into station systems, and align maintenance strategy with asset criticality and project budget.

Conclusion

These website papers are structured to demonstrate how CESC approaches engineering subjects at system level, not only at equipment catalogue level. The intended message to clients is that CESC can define studies, design criteria, scheme

architecture, implementation strategy, testing approach, and brownfield execution method across protection, control, digital integration, retrofit, and asset management scopes.